

SDS 264: Mini Project 1

Guts Are Sour When You're Sad and in Love!: A Text Analysis of Olivia Rodrigo's Existing Albums

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Introduction

As of 2026, Olivia has released three stellar full-length albums, including “SOUR” in 2021, “GUTS” in 2023, and “you seem pretty sad for a girl so in love” (YOUSEEMSAD, abbreviated in this article) in 2026. Each album (excluding deluxe versions) includes more songs than the last, going from 11 songs in “SOUR,” to 12 in “GUTS,” to 13 tracks in YOUSEEMSAD! Her albums paint a picture of heartbreak, jealousy, anger, and acceptance. In this analysis, we will first examine the defining genres and tags of her music, and then delve deeper, analyzing the common themes in her lyrics, and how they have changed over the course of her music career.

All data scraped from the website Genius using the lyricsr package.

First, let's take a look at when each album released singles. The following table shows all of the release dates from each album! As you can see, her most recent album only contained two lead singles prior to the full release of the album, while her first two albums had three lead singles!

```
# A tibble: 10 x 2
# Groups:   album [3]
  album                      release_date1
  <chr>                      <chr>
1 SOUR                      01-08-2021
2 SOUR                      04-01-2021
3 SOUR                      05-14-2021
4 SOUR                      05-21-2021
5 GUTS                      06-30-2023
```

6	GUTS	09-08-2023
7	GUTS	08-11-2023
8	you seem pretty sad for a girl so in love	05-22-2026
9	you seem pretty sad for a girl so in love	04-17-2026
10	you seem pretty sad for a girl so in love	06-12-2026

Genres and Tags

A helpful feature of the Genius website is that each song is given a number of tags! The following table shows the most common words used in the tags throughout Olivia's discography. Clearly, her music is most aligned with tags that have to do with pop music, however, she still has quite a few tags involving rock music! While some might see Olivia as a glittery purple popstar, she's got some grit to her sound as well.

Common Tags

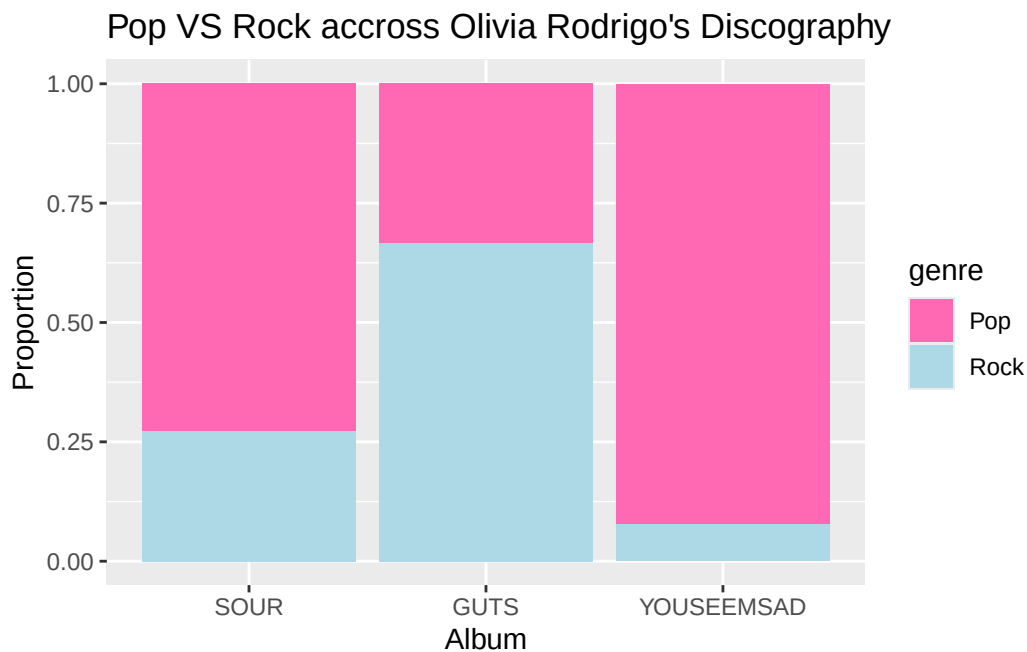
	[,1]
Number of Pop Tags	201
Number of Rock Tags	116
Number of Alternative Tags	88
Number of Folk Tags	41
Number of Electric Tags	9

However, what about the genre classification that Genius gives each song? The following plot shows the proportion of rock and pop songs between Olivia's three albums. Her second album is a clear outlier from her first and third! But while Olivia got to rock out in her GUTS "era," she went back to her pop princess roots for her third studio album. "My Way" is the *one* rock song in Olivia's most recent album!

```
#| warning: FALSE
#| message: FALSE
#| echo: FALSE

songs_olivia2 |>
  ggplot(aes
    (x = fct_reorder(album, year),
     fill = genre)
  ) +
  geom_bar(
    position = "fill"
  ) +
```

```
labs(
  x = "Album",
  y = "Proportion",
  title = "Pop VS Rock accross Olivia Rodrigo's Discography"
) +
scale_x_discrete(
  labels =
    c("you seem pretty sad for a girl so in love" = "YOUSEEMSAD")
) +
scale_fill_manual(
  values = c("Pop" = "hotpink",
             "Rock" = "lightblue"))
```



#In this code chunk, I created a simple bar graph showing the proportion of rock vs pop songs

But now it's time to dive into the lyrics!! More specifically, the *words* in the lyrics.

Common Words

Well, maybe not words quite yet... The following table shows the most common vocalizing “words” present in Olivia Rodrigo’s lyrics. As we can see, she loves to sing “AH!” almost as much as she likes to sing about love!

Vocalizing

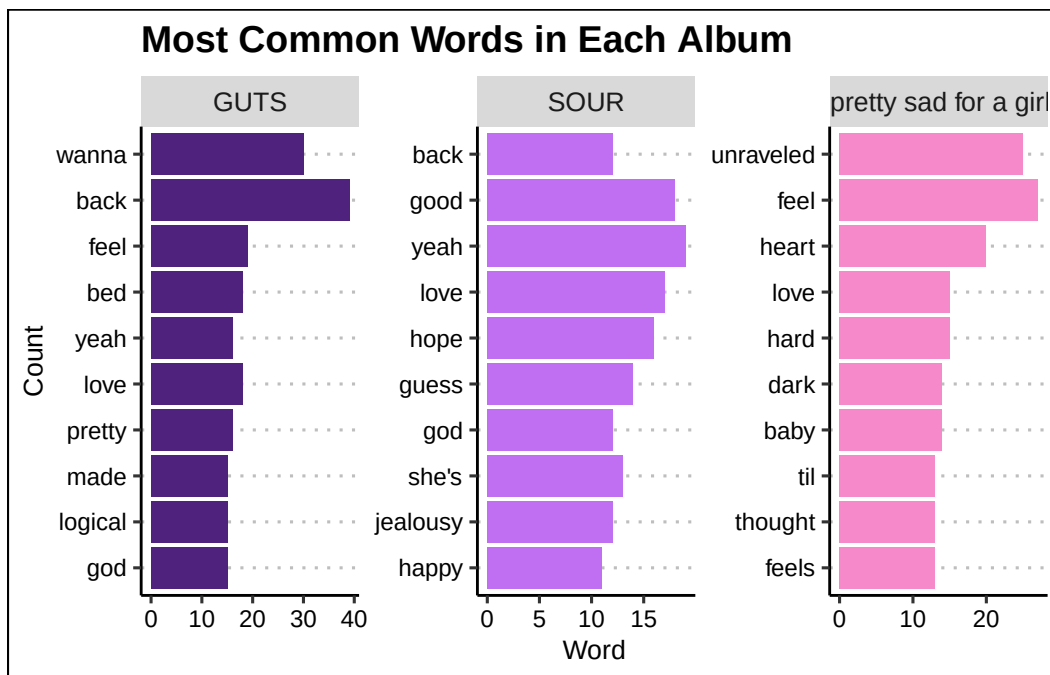
A tibble: 7 x 2

	word	n
	<chr>	<int>
1	ah	100
2	ooh	40
3	la	24
4	ha	15
5	huh	2
6	ugh	1
7	uh	1

But what *actual* words does Olivia like to sing... The following plot shows the ten most common words in each album... and the *one* word that's in the top ten for each album is...drumroll please...LOVE!! Honorable mentions include “feel,” which is on the list for GUTS and YOUSEEMSAD, and “god,” which is on the list for both SOUR and GUTS.

This lists do all contain a common theme, however, and it is that of emotions! Olivia's music mainly has to do with the complex feelings she has around love, self-image, and heartbreak.

Let's find out if those feelings are positive or negative...



Sentiment

Most words, believe it or not, can be classified into either “positive” or “negative.” The following word cloud shows the most common positive words from Olivia’s discography.

Positive WordCloud



As we can see, “love” and “good” are pretty high up there. Some interesting words on there, however, are words such as “logical,” and “cure.” One would think that those words would be positive, but the way Olivia uses them in her songs named “Logical” and “The Cure,” are anything but. These songs are deeply emotional and heart wrenching!

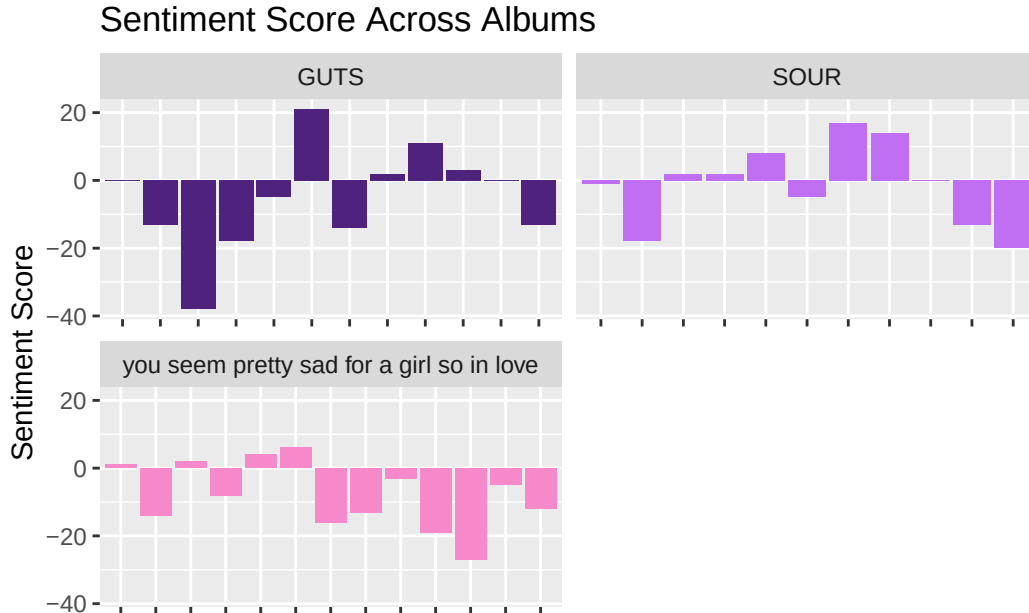
Let's take a look at the wordcloud for the most common negative words!

Negative WordCloud



Lots of curse words in this one... sorry... But as we can see, “unraveled” is the most prevalent negative word, with “wrong” and “hard” coming close behind. This one also contains many negative *emotions*, like jealousy, crazy, sick, doubt, and insane.

However, what if we could visualize the net sentiment for a single song? By counting the amount of positive and negative words in each song, and then finding the difference, we can visualize the distribution of net sentiment across songs in each album! Check it out!



This plot shows us that perhaps our negative wordcloud had a lot more words to pick from... It is very interesting though to see how her albums have somewhat of a flow to them when it comes to their average sentiment!

Conclusion

Through analysis of the data provided by Genius, we can conclude that Olivia Rodrigo is mainly a pop artist with a couple solid rock hits. Her music has a distinct theme of love, relationships, and feelings. Teenage girls everywhere can certainly empathize with her gritty, sorrowful, and honest music.